Xiao (Griffin) Wang

UMN Workshop on Relative Langlands

June 6, 2024

Kashiwara Crystals in Endoscopy

Fundamental Lemma made Fun-to-handle (tables included?)

Introduction

- In my recent work (arXiv:2402.19331), I gave a new proof of the fundamental lemma for spherical Hecke algebras, using multiplicative Hitchin fibrations.

Introduction

- In my recent work (arXiv:2402.19331), I gave a new proof of the fundamental lemma for spherical Hecke algebras, using multiplicative Hitchin fibrations.
- Compared to the Lie algebra version proved by Ngô (2010), there are many interesting new features.

Introduction

- In my recent work (arXiv:2402.19331), I gave a new proof of the fundamental lemma for spherical Hecke algebras, using multiplicative Hitchin fibrations.
- Compared to the Lie algebra version proved by Ngô (2010), there are many interesting new features.
- The goal of this talk is to introduce one of those features, called the asymptotic fundamental lemma (AsFL), and explain why it may be promising to study in a broader context.

Outline of the Talk

1. Significance of FL in trace formula, using SL_2 as an example

Outline of the Talk

1. Significance of FL in trace formula, using SL_2 as an example
2. FL and AsFL

Outline of the Talk

1. Significance of FL in trace formula, using SL_2 as an example
2. FL and AsFL
3. Transfer factor

Outline of the Talk

1. Significance of FL in trace formula, using SL_2 as an example
2. FL and AsFL
3. Transfer factor
4. Kashiwara crystals and some open questions

Outline of the Talk

1. Significance of FL in trace formula, using SL_2 as an example
2. FL and AsFL
3. Transfer factor
4. Kashiwara crystals and some open questions

Not in this talk

Outline of the Talk

1. Significance of FL in trace formula, using SL_2 as an example
2. FL and AsFL
3. Transfer factor
4. Kashiwara crystals and some open questions

Not in this talk

- The proof of FL itself, outside some necessary ideas.

Outline of the Talk

1. Significance of FL in trace formula, using SL_2 as an example
2. FL and AsFL
3. Transfer factor
4. Kashiwara crystals and some open questions

Not in this talk

- The proof of FL itself, outside some necessary ideas.
- Multiplicative Hitchin fibrations.

Motivating example: SL_2

1 Motivating example: SL_2
2 From FL to AsFL
3 On Top of Mindolluin
4 Asymptotic Fundamental Lemma
5 The Transfer Factor
6 Kashiwara Crystals

- Let $G = \mathrm{SL}_2$.

Langlands Functoriality

- Let $G = \mathrm{SL}_2$.
- Let $H = \mathrm{SO}_2$ associated with standard quadratic form $x^2 + y^2$.

Langlands Functoriality

- Let $G = \mathrm{SL}_2$.
- Let $H = \mathrm{SO}_2$ associated with standard quadratic form $x^2 + y^2$.
- Let θ be a (quasi-)character of $[H] \coloneqq H(\mathbb{Q})\backslash H(\mathbf{A}_{\mathbb{Q}})$:

$$\theta \colon [H] \longrightarrow \mathbb{C}^\times.$$

Langlands Functoriality

- Let $G = \mathrm{SL}_2$.
- Let $H = \mathrm{SO}_2$ associated with standard quadratic form $x^2 + y^2$.
- Let θ be a (quasi-)character of $[H] \coloneqq H(\mathbb{Q})\backslash H(\mathbf{A}_{\mathbb{Q}})$:

$$\theta \colon [H] \longrightarrow \mathbb{C}^\times.$$

- In the introduction of *Les Débuts d'une Formule Traces Stables*, Langlands demonstrates the existence of an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is a "functorial lift" of θ.

Langlands Functoriality

- Let $G = \mathrm{SL}_2$.
- Let $H = \mathrm{SO}_2$ associated with standard quadratic form $x^2 + y^2$.
- Let θ be a (quasi-)character of $[H] \coloneqq H(\mathbb{Q})\backslash H(\mathbf{A}_{\mathbb{Q}})$:

$$\theta \colon [H] \longrightarrow \mathbb{C}^\times.$$

- In the introduction of *Les Débuts d'une Formule Traces Stables*, Langlands demonstrates the existence of an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is a "functorial lift" of θ.
- The argument uses stable trace formula and a primitive form of fundamental lemma for $G = \mathrm{SL}_2$.

Langlands Functoriality

- Let $G = \mathrm{SL}_2$.
- Let $H = \mathrm{SO}_2$ associated with standard quadratic form $x^2 + y^2$.
- Let θ be a (quasi-)character of $[H] \coloneqq H(\mathbb{Q})\backslash H(\mathbf{A}_{\mathbb{Q}})$:

$$\theta \colon [H] \longrightarrow \mathbb{C}^\times.$$

- In the introduction of *Les Débuts d'une Formule Traces Stables*, Langlands demonstrates the existence of an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is a "functorial lift" of θ.
- The argument uses stable trace formula and a primitive form of fundamental lemma for $G = \mathrm{SL}_2$.
- I will only give a rough outline.

- We restrict to local field $\mathbb{Q}_p$ ($p \neq 2$), and suppose $\sqrt{-1} \notin \mathbb{Q}_p$.

Endoscopy for SL_2

- We restrict to local field $\mathbb{Q}_p$ ($p \neq 2$), and suppose $\sqrt{-1} \notin \mathbb{Q}_p$.
- Let $\gamma, \gamma' \in G(\mathbb{Q}_p)$

$$\gamma = \frac{1}{\sqrt{1+p^2}} \begin{pmatrix} 1 & -p \\ p & 1 \end{pmatrix}, \quad \gamma' = \frac{1}{\sqrt{1+p^2}} \begin{pmatrix} 1 & -1 \\ p^2 & 1 \end{pmatrix}.$$

Endoscopy for SL_2

- We restrict to local field $\mathbb{Q}_p$ ($p \neq 2$), and suppose $\sqrt{-1} \notin \mathbb{Q}_p$.
- Let $\gamma, \gamma' \in G(\mathbb{Q}_p)$

$$\gamma = \frac{1}{\sqrt{1+p^2}} \begin{pmatrix} 1 & -p \\ p & 1 \end{pmatrix}, \quad \gamma' = \frac{1}{\sqrt{1+p^2}} \begin{pmatrix} 1 & -1 \\ p^2 & 1 \end{pmatrix}.$$

- Let $\gamma_H = \gamma$, regarded as an element of H.

Endoscopy for SL_2

- We restrict to local field $\mathbb{Q}_p$ ($p \neq 2$), and suppose $\sqrt{-1} \notin \mathbb{Q}_p$.
- Let $\gamma, \gamma' \in G(\mathbb{Q}_p)$

$$\gamma = \frac{1}{\sqrt{1+p^2}} \begin{pmatrix} 1 & -p \\ p & 1 \end{pmatrix}, \quad \gamma' = \frac{1}{\sqrt{1+p^2}} \begin{pmatrix} 1 & -1 \\ p^2 & 1 \end{pmatrix}.$$

- Let $\gamma_H = \gamma$, regarded as an element of H.
- γ and γ' are $G(\overline{\mathbb{Q}}_p)$-conjugate but not $G(\mathbb{Q}_p)$-conjugate: the *endoscopy* phenomenon.

Example of Fundamental Lemma

- Consider three sets of $\mathbb{Z}_p$-lattices in $\mathbb{Q}_p^2$:

$$\begin{aligned}
\mathcal{M}_\gamma &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \gamma L \subset L \right\}, \\
\mathcal{M}_{\gamma'} &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \gamma' L \subset L \right\}, \\
\mathcal{M}_{H,\gamma_H} &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \exists h \in H(\mathbb{Q}_p),\ \text{s.t. } L = h\mathbb{Z}_p^2 \text{ and } \gamma_H L \subset L \right\}.
\end{aligned}$$

Example of Fundamental Lemma

- Consider three sets of $\mathbb{Z}_p$-lattices in $\mathbb{Q}_p^2$:

$$\begin{aligned}
\mathcal{M}_\gamma &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \gamma L \subset L \right\}, \\
\mathcal{M}_{\gamma'} &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \gamma' L \subset L \right\}, \\
\mathcal{M}_{H,\gamma_H} &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \exists h \in H(\mathbb{Q}_p),\ \text{s.t. } L = h\mathbb{Z}_p^2 \text{ and } \gamma_H L \subset L \right\}.
\end{aligned}$$

- In this case, the fundamental lemma claims an equality of lattice counts

$$-p^{-1}\left[\#\mathcal{M}_\gamma - \#\mathcal{M}_{\gamma'}\right] = \#\mathcal{M}_{H,\gamma_H}.$$

Example of Fundamental Lemma

- Consider three sets of $\mathbb{Z}_p$-lattices in $\mathbb{Q}_p^2$:

$$\begin{aligned}\mathcal{M}_\gamma &= \left\{L \subset \mathbb{Q}_p^2 \,\middle|\, \gamma L \subset L\right\},\\ \mathcal{M}_{\gamma'} &= \left\{L \subset \mathbb{Q}_p^2 \,\middle|\, \gamma' L \subset L\right\},\\ \mathcal{M}_{H,\gamma_H} &= \left\{L \subset \mathbb{Q}_p^2 \,\middle|\, \exists h \in H(\mathbb{Q}_p),\ \text{s.t. } L = h\mathbb{Z}_p^2 \text{ and } \gamma_H L \subset L\right\}.\end{aligned}$$

- In this case, the fundamental lemma claims an equality of lattice counts

$$-p^{-1}\left[\#\mathcal{M}_\gamma - \#\mathcal{M}_{\gamma'}\right] = \#\mathcal{M}_{H,\gamma_H}.$$

- Rewritten using orbital integrals:

$$-p^{-1}\left[\mathbf{O}_\gamma(\mathbf{1}_{G(\mathbb{Z}_p)}) - \mathbf{O}_{\gamma'}(\mathbf{1}_{G(\mathbb{Z}_p)})\right] = \mathbf{O}_{H,\gamma_H}(\mathbf{1}_{H(\mathbb{Z}_p)}).$$

Example of Fundamental Lemma

- Consider three sets of $\mathbb{Z}_p$-lattices in $\mathbb{Q}_p^2$:

$$\begin{aligned}
\mathcal{M}_\gamma &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \gamma L \subset L \right\}, \\
\mathcal{M}_{\gamma'} &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \gamma' L \subset L \right\}, \\
\mathcal{M}_{H,\gamma_H} &= \left\{ L \subset \mathbb{Q}_p^2 \;\middle|\; \exists h \in H(\mathbb{Q}_p),\ \text{s.t. } L = h\mathbb{Z}_p^2 \text{ and } \gamma_H L \subset L \right\}.
\end{aligned}$$

- In this case, the fundamental lemma claims an equality of lattice counts

$$-p^{-1}\left[\#\mathcal{M}_\gamma - \#\mathcal{M}_{\gamma'}\right] = \#\mathcal{M}_{H,\gamma_H}.$$

- Rewritten using orbital integrals:

$$-p^{-1}\left[\mathbf{O}_\gamma(\mathbf{1}_{G(\mathbb{Z}_p)}) - \mathbf{O}_{\gamma'}(\mathbf{1}_{G(\mathbb{Z}_p)})\right] = \mathbf{O}_{H,\gamma_H}(\mathbf{1}_{H(\mathbb{Z}_p)}).$$

- There are Lie algebra analogues as well.

Trace Formula and Stability

- Let $f \in C_c^\infty(G(\mathbf{A}_\mathbb{Q}))$ be a test function, then the trace of f on $L^2([G])$ (where $[G] \coloneqq G(\mathbb{Q})\backslash G(\mathbf{A}_\mathbb{Q})$) *morally* decomposes as

$$\mathrm{Tr}(f) \approx \underbrace{\sum_{\pi \in \mathrm{Irrep}(G(\mathbf{A}_\mathbb{Q}))} m_\pi \, \mathrm{Tr}_\pi(f)}_{\text{(characters)}} \approx \underbrace{\sum_{\gamma \in G(\mathbb{Q})/\sim} a_\gamma \, \mathbf{O}_\gamma(f)}_{\text{(class functions)}} .$$

Trace Formula and Stability

- Let $f \in C_c^\infty(G(\mathbf{A}_\mathbb{Q}))$ be a test function, then the trace of f on $L^2([G])$ (where $[G] \coloneqq G(\mathbb{Q})\backslash G(\mathbf{A}_\mathbb{Q})$) *morally* decomposes as

$$\mathrm{Tr}(f) \approx \underbrace{\sum_{\pi \in \mathrm{Irrep}(G(\mathbf{A}_\mathbb{Q}))} m_\pi \, \mathrm{Tr}_\pi(f)}_{\text{(characters)}} \approx \underbrace{\sum_{\gamma \in G(\mathbb{Q})/\sim} a_\gamma \, \mathbf{O}_\gamma(f)}_{\text{(class functions)}} .$$

- *Trace formula:* a rigorous equality using a meaningfully modified summation $\sum'$.

Trace Formula and Stability

- Let $f \in C_c^\infty(G(\mathbf{A}_\mathbb{Q}))$ be a test function, then the trace of f on $L^2([G])$ (where $[G] := G(\mathbb{Q})\backslash G(\mathbf{A}_\mathbb{Q})$) *morally* decomposes as

$$\mathrm{Tr}(f) \approx \underbrace{\sum_{\pi \in \mathrm{Irrep}(G(\mathbf{A}_\mathbb{Q}))} m_\pi \, \mathrm{Tr}_\pi(f)}_{\text{(characters)}} \approx \underbrace{\sum_{\gamma \in G(\mathbb{Q})/\sim} a_\gamma \, \mathbf{O}_\gamma(f)}_{\text{(class functions)}} .$$

- *Trace formula:* a rigorous equality using a meaningfully modified summation $\sum'$.
- The trace Tr is *not stable*: if $\gamma, \gamma' \in G(\mathbf{A}_\mathbb{Q})$ have the same characteristic polynomial everywhere, then $\mathbf{O}_\gamma$ and $\mathbf{O}_{\gamma'}$ may have different contribution to Tr:

$$\mathrm{Tr}(f) \neq \mathrm{STr}(f) := \sum_\gamma \left[b_\gamma \sum_{\gamma' \sim \gamma} \mathbf{O}_{\gamma'}(f) \right],$$

where STr is the stable trace by forcing $\mathbf{O}_\gamma$ and $\mathbf{O}_{\gamma'}$ to have the same contribution.

Trace Formula and Stability

- Let $f \in C_c^\infty(G(\mathbf{A}_\mathbb{Q}))$ be a test function, then the trace of f on $L^2([G])$ (where $[G] := G(\mathbb{Q})\backslash G(\mathbf{A}_\mathbb{Q})$) *morally* decomposes as

$$\mathrm{Tr}(f) \approx \underbrace{\sum_{\pi \in \mathrm{Irrep}(G(\mathbf{A}_\mathbb{Q}))} m_\pi \, \mathrm{Tr}_\pi(f)}_{\text{(characters)}} \approx \underbrace{\sum_{\gamma \in G(\mathbb{Q})/\sim} a_\gamma \, \mathbf{O}_\gamma(f)}_{\text{(class functions)}} .$$

- *Trace formula:* a rigorous equality using a meaningfully modified summation $\sum'$.
- The trace Tr is *not stable*: if $\gamma, \gamma' \in G(\mathbf{A}_\mathbb{Q})$ have the same characteristic polynomial everywhere, then $\mathbf{O}_\gamma$ and $\mathbf{O}_{\gamma'}$ may have different contribution to Tr:

$$\mathrm{Tr}(f) \neq \mathrm{STr}(f) := \sum_\gamma \left[b_\gamma \sum_{\gamma' \sim \gamma} \mathbf{O}_{\gamma'}(f) \right],$$

  where STr is the stable trace by forcing $\mathbf{O}_\gamma$ and $\mathbf{O}_{\gamma'}$ to have the same contribution.
- Instability causes many problems in global setting, but it is also a blessing in disguise.

Stable Trace Formula

- Langlands' idea was to write the error term $\mathrm{Tr}(f) - \mathrm{STr}(f)$ as

$$\mathrm{Tr}(f) - \mathrm{STr}(f) = \sum_{G'} c_{G'} \, \mathrm{STr}_{G'}(f_{G'})$$

for various G' (including H) smaller than G and some number $c_{G'} \in \mathbb{C}$.

Stable Trace Formula

- Langlands' idea was to write the error term $\mathrm{Tr}(f) - \mathrm{STr}(f)$ as

$$\mathrm{Tr}(f) - \mathrm{STr}(f) = \sum_{G'} c_{G'} \, \mathrm{STr}_{G'}(f_{G'})$$

  for various G' (including H) smaller than G and some number $c_{G'} \in \mathbb{C}$.
- The functions $f_{G'}$ are the *transfers* of f. They are compatible in the following sense:

$$\begin{aligned} \text{compatible in the sense of Functoriality} &= \text{compatible in the sense of FL} \\ &= \text{compatible in the sense of Transfer} \\ &= \text{compatible with Satake when unramified.} \end{aligned}$$

Stable Trace Formula

- Langlands' idea was to write the error term $\mathrm{Tr}(f) - \mathrm{STr}(f)$ as

$$\mathrm{Tr}(f) - \mathrm{STr}(f) = \sum_{G'} c_{G'} \, \mathrm{STr}_{G'}(f_{G'})$$

  for various G' (including H) smaller than G and some number $c_{G'} \in \mathbb{C}$.
- The functions $f_{G'}$ are the *transfers* of f. They are compatible in the following sense:

$$\begin{aligned} \text{compatible in the sense of Functoriality} &= \text{compatible in the sense of FL} \\ &= \text{compatible in the sense of Transfer} \\ &= \text{compatible with Satake when unramified.} \end{aligned}$$

- This process is called *stabilization*. It is proven by using fundamental lemma on the geometric (i.e., orbital integral) side of Tr.

Back to Functoriality

- Reminder: $G = \mathrm{SL}_2$, $H = \mathrm{SO}_2$.

Back to Functoriality

- Reminder: $G = \mathrm{SL}_2$, $H = \mathrm{SO}_2$.
- θ a quasi-character of $[H]$.

Back to Functoriality

- Reminder: $G = \mathrm{SL}_2$, $H = \mathrm{SO}_2$.
- θ a quasi-character of $[H]$.
- Want: an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is *compatible* with θ.

Back to Functoriality

- Reminder: $G = \mathrm{SL}_2$, $H = \mathrm{SO}_2$.
- θ a quasi-character of $[H]$.
- Want: an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is *compatible* with θ.
- By carefully choosing test function f, one can make $\mathrm{STr}(f) = 0$, and so

$$\mathrm{Tr}(f) = \mathrm{STr}(f) + \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{Tr}_{G'}(f_{G'}),$$

where in this case G' are various forms of SO_2, hence abelian.

Back to Functoriality

- Reminder: $G = \mathrm{SL}_2$, $H = \mathrm{SO}_2$.
- θ a quasi-character of $[H]$.
- Want: an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is *compatible* with θ.
- By carefully choosing test function f, one can make $\mathrm{STr}(f) = 0$, and so

$$\mathrm{Tr}(f) = \mathrm{STr}(f) + \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{Tr}_{G'}(f_{G'}),$$

  where in this case G' are various forms of SO_2, hence abelian.
- Choosing f more carefully so that the contribution from $\mathrm{Tr}_\theta(f_H)$ is non-vanishing.

Back to Functoriality

- Reminder: $G = \mathrm{SL}_2$, $H = \mathrm{SO}_2$.
- θ a quasi-character of $[H]$.
- Want: an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is *compatible* with θ.
- By carefully choosing test function f, one can make $\mathrm{STr}(f) = 0$, and so

$$\mathrm{Tr}(f) = \mathrm{STr}(f) + \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{Tr}_{G'}(f_{G'}),$$

  where in this case G' are various forms of SO_2, hence abelian.
- Choosing f more carefully so that the contribution from $\mathrm{Tr}_\theta(f_H)$ is non-vanishing.
- Then the corresponding term in $\mathrm{Tr}(f)$ that is compatible with θ must also be non-vanishing. Therefore a functorial lift π must exist.

Back to Functoriality

- Reminder: $G = \mathrm{SL}_2$, $H = \mathrm{SO}_2$.
- θ a quasi-character of $[H]$.
- Want: an automorphic representation π of $G(\mathbf{A}_{\mathbb{Q}})$ that is *compatible* with θ.
- By carefully choosing test function f, one can make $\mathrm{STr}(f) = 0$, and so

$$\mathrm{Tr}(f) = \mathrm{STr}(f) + \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{STr}_{G'}(f_{G'}) = \sum_{G'} c_{G'}\,\mathrm{Tr}_{G'}(f_{G'}),$$

  where in this case G' are various forms of SO_2, hence abelian.
- Choosing f more carefully so that the contribution from $\mathrm{Tr}_\theta(f_H)$ is non-vanishing.
- Then the corresponding term in $\mathrm{Tr}(f)$ that is compatible with θ must also be non-vanishing. Therefore a functorial lift π must exist.

Summary

Fundamental lemma implies functoriality for pairs (G, H) associated with instability of trace formula (equivalently, endoscopy).

From FL to AsFL

1 Motivating example: SL_2
2 From FL to AsFL
3 On Top of Mindolluin
4 Asymptotic Fundamental Lemma
5 The Transfer Factor
6 Kashiwara Crystals

Statement of Lie-FL

Statement of Lie-FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.

Statement of Lie-FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$, $\mathfrak{g}$ its Lie algebra.

Statement of Lie-FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$, $\mathfrak{g}$ its Lie algebra.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.

Statement of Lie-FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$, $\mathfrak{g}$ its Lie algebra.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G, $\mathfrak{h}$ the Lie algebra of H.

Statement of Lie-FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$, $\mathfrak{g}$ its Lie algebra.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G, $\mathfrak{h}$ the Lie algebra of H.

Theorem (Ngô, 2010)

For any matching G-regular semisimple pairs $(X_H, X) \in \mathfrak{h}(F) \times \mathfrak{g}(F)$, we have equality in orbital integrals

$$\mathbf{SO}_{H,X_H}(\mathbf{1}_{\mathfrak{h}(\mathcal{O})}, \mathrm{d}t) = \Delta_0^{\mathfrak{g}}(X_H, X)\, \mathbf{O}_X^{\kappa}(\mathbf{1}_{\mathfrak{g}(\mathcal{O})}, \mathrm{d}t),$$

where $\Delta_0^{\mathfrak{g}}(X_H, X)$ is the appropriately defined transfer factor.

Statement of Lie-FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$, $\mathfrak{g}$ its Lie algebra.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G, $\mathfrak{h}$ the Lie algebra of H.

Theorem (Ngô, 2010)

For any matching G-regular semisimple pairs $(X_H, X) \in \mathfrak{h}(F) \times \mathfrak{g}(F)$, we have equality in orbital integrals

$$\mathbf{SO}_{H,X_H}(\mathbf{1}_{\mathfrak{h}(\mathcal{O})}, \mathrm{d}t) = \Delta_0^{\mathfrak{g}}(X_H, X)\, \mathbf{O}_X^{\kappa}(\mathbf{1}_{\mathfrak{g}(\mathcal{O})}, \mathrm{d}t),$$

where $\Delta_0^{\mathfrak{g}}(X_H, X)$ is the appropriately defined transfer factor.

- Admissible embedding $\xi\colon {}^L H \to {}^L G$ has (almost) no role here.

Statement of Lie-FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$, $\mathfrak{g}$ its Lie algebra.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G, $\mathfrak{h}$ the Lie algebra of H.

Theorem (Ngô, 2010)

For any matching G-regular semisimple pairs $(X_H, X) \in \mathfrak{h}(F) \times \mathfrak{g}(F)$, we have equality in orbital integrals

$$\mathbf{SO}_{H,X_H}(\mathbf{1}_{\mathfrak{h}(\mathcal{O})}, \mathrm{d}t) = \Delta_0^{\mathfrak{g}}(X_H, X)\, \mathbf{O}_X^{\kappa}(\mathbf{1}_{\mathfrak{g}(\mathcal{O})}, \mathrm{d}t),$$

where $\Delta_0^{\mathfrak{g}}(X_H, X)$ is the appropriately defined transfer factor.

- Admissible embedding $\xi\colon {}^L H \to {}^L G$ has (almost) no role here.
- The functions $\mathbf{1}_{\mathfrak{h}(\mathcal{O})}$ and $\mathbf{1}_{\mathfrak{g}(\mathcal{O})}$ are fixed.

Statement of FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G.

Statement of FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G.

Theorem (W 2024)

For any matching strongly G-regular semisimple pairs $(\gamma_H, \gamma) \in H(F) \times G(F)$ and spherical function f on $G(F)$, we have equality in orbital integrals

$$\mathbf{SO}_{H,\gamma_H}(f_{H,\xi}, \mathrm{d}t) = \Delta_0(\gamma_H, \gamma)\, \mathbf{O}^{\kappa}_{\gamma}(f, \mathrm{d}t),$$

where $f_{H,\xi}$ is the spherical transfer of f via ξ, and $\Delta_0(\gamma_H, \gamma)$ is the appropriately defined transfer factor.

Statement of FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G.

Theorem (W 2024)

For any matching strongly G-regular semisimple pairs $(\gamma_H, \gamma) \in H(F) \times G(F)$ and spherical function f on $G(F)$, we have equality in orbital integrals

$$\mathbf{SO}_{H,\gamma_H}(f_{H,\xi}, \mathrm{d}t) = \Delta_0(\gamma_H, \gamma)\, \mathbf{O}^{\kappa}_{\gamma}(f, \mathrm{d}t),$$

where $f_{H,\xi}$ is the spherical transfer of f via ξ, and $\Delta_0(\gamma_H, \gamma)$ is the appropriately defined transfer factor.

- Admissible embedding $\xi\colon {}^L H \to {}^L G$ is used.

Statement of FL

- Let $k = \mathbb{F}_q$, $\mathcal{O} = k[[\pi]]$, $F = k((\pi))$.
- Let G be a reductive group scheme over $\mathcal{O}$.
- Suppose $\mathrm{char}(k)$ is larger than twice the Coxeter number of G.
- Let (H, κ, ξ) be an unramified endoscopic datum of G.

Theorem (W 2024)

For any matching strongly G-regular semisimple pairs $(\gamma_H, \gamma) \in H(F) \times G(F)$ and spherical function f on $G(F)$, we have equality in orbital integrals

$$\mathbf{SO}_{H,\gamma_H}(f_{H,\xi}, \mathrm{d}t) = \Delta_0(\gamma_H, \gamma)\, \mathbf{O}^{\kappa}_{\gamma}(f, \mathrm{d}t),$$

where $f_{H,\xi}$ is the spherical transfer of f via ξ, and $\Delta_0(\gamma_H, \gamma)$ is the appropriately defined transfer factor.

- Admissible embedding $\xi\colon {}^L H \to {}^L G$ is used.
- The function f can be any spherical function (not just $\mathbf{1}_{G(\mathcal{O})}$).

The Idea of Proving (Lie)-FL

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are “limits” of simple ones.

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The group case

The Idea of Proving (Lie)-FL

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The group case

- The multiplicative Hitchin fibration also encodes orbital integrals into families.

The Idea of Proving (Lie)-FL

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The group case

- The multiplicative Hitchin fibration also encodes orbital integrals into families.
- Note that it encodes both conjugacy class γ and spherical function f.

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The group case

- The multiplicative Hitchin fibration also encodes orbital integrals into families.
- Note that it encodes both conjugacy class γ and spherical function f.
- Mimicking Ngô's argument, one reduces the problem to some "simple cases".

The Idea of Proving (Lie)-FL

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The group case

- The multiplicative Hitchin fibration also encodes orbital integrals into families.
- Note that it encodes both conjugacy class γ and spherical function f.
- Mimicking Ngô's argument, one reduces the problem to some "simple cases".
- "Simple case" here means simple γ and f, but there are more than one f possible.

The Idea of Proving (Lie)-FL

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The group case

- The multiplicative Hitchin fibration also encodes orbital integrals into families.
- Note that it encodes both conjugacy class γ and spherical function f.
- Mimicking Ngô's argument, one reduces the problem to some "simple cases".
- "Simple case" here means simple γ and f, but there are more than one f possible.
- As a result, there are additional "simple cases" given by those $f \neq \mathbf{1}_{G(\mathcal{O})}$.

The Idea of Proving (Lie)-FL

The Lie algebra case

- Hitchin fibration encodes orbital integrals (for Lie algebras) into an algebraic family.
- By some ingenious work, Ngô shows that complicated orbital integrals are "limits" of simple ones.
- Those "simple cases" for Lie-FL turn out to be either trivial, or isomorphic to the specific case where $G = \mathrm{SL}_2$ and $H = \mathrm{SO}_2$ discussed before, all computable by hand.

The group case

- The multiplicative Hitchin fibration also encodes orbital integrals into families.
- Note that it encodes both conjugacy class γ and spherical function f.
- Mimicking Ngô's argument, one reduces the problem to some "simple cases".
- "Simple case" here means simple γ and f, but there are more than one f possible.
- As a result, there are additional "simple cases" given by those $f \neq \mathbf{1}_{G(\mathcal{O})}$.
- These cases lead to the *asymptotic fundamental lemma*.

On Top of Mindolluin

1 Motivating example: SL_2
2 From FL to AsFL
3 On Top of Mindolluin
4 Asymptotic Fundamental Lemma
5 The Transfer Factor
6 Kashiwara Crystals

$$
\begin{tikzcd}
\boxed{\mathbf{O}_\gamma(f)} \arrow[r, squiggly, "\Delta_{\mathrm{III}_1}"] & \overset{G\text{-side}}{\mathbf{O}^\kappa_\gamma(f)} \arrow[r, dash, "\Delta_0=\Delta_{\mathrm{I}}\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_2}\Delta_{\mathrm{IV}}", "\mathbf{FL}"'] \arrow[d, squiggly, "\mathrm{MASF}"'] & \overset{H\text{-side}}{\mathbf{SO}_{H,\gamma_H}(f_{H,\xi})} \arrow[d, squiggly, "\text{MASF for } H"] \\
\boxed{\begin{array}{c} W \rtimes \sigma \\ \text{v.s.} \\ \mathrm{N}_G(T) \rtimes \sigma \end{array}} \arrow[r, dashed, "\Delta_{\mathrm{I}}"] & \#_f\left[\mathcal{M}_\gamma/\check{\mathbb{X}}\right]_\kappa \arrow[r, dashed] \arrow[d, squiggly, "\text{mH-fibrations}"'] & \#_{f_{H,\xi}}\left[\mathcal{M}_{H,\gamma_H}/\check{\mathbb{X}}\right]_{\mathrm{st}} \arrow[d, squiggly, "\text{mH-fibrations for } H"] \\
\boxed{\text{Tate twists}} \arrow[r, dashed, "\Delta_{\mathrm{IV}}"] & \mathrm{Irr}_f\left[\mathcal{M}_\gamma/\check{\mathbb{X}}\right]_\kappa \arrow[r, dashed] \arrow[d, squiggly, "\check{\mathbf{G}}\text{-crystals}"'] & \mathrm{Irr}_{f_{H,\xi}}\left[\mathcal{M}_{H,\gamma_H}/\check{\mathbb{X}}\right]_{\mathrm{st}} \arrow[d, squiggly, "\check{\mathbf{H}}\text{-crystals}"] \\
 & \mathfrak{V}[\nu]_\kappa \arrow[r, dash, "\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_2}", "\mathbf{AsFL}"'] & \mathfrak{V}_H[\nu]_{\mathrm{st}}
\end{tikzcd}
$$

Asymptotic Fundamental Lemma

1 Motivating example: SL_2
2 From FL to AsFL
3 On Top of Mindolluin
4 Asymptotic Fundamental Lemma
5 The Transfer Factor
6 Kashiwara Crystals

A Special Case: SL_3

- Let $G = \mathrm{SL}_3$ and ${}^L G = \mathrm{PGL}_3(\mathbb{C})$.

A Special Case: SL_3

- Let $G = \mathrm{SL}_3$ and ${}^L G = \mathrm{PGL}_3(\mathbb{C})$.
- Let $\kappa \in {}^L G$ be the element represented by

$$\kappa = \begin{pmatrix} 1 & & \\ & \zeta_3 & \\ & & \zeta_3^2 \end{pmatrix}, \quad \zeta_3^3 = 1.$$

A Special Case: SL_3

- Let $G = \mathrm{SL}_3$ and ${}^L G = \mathrm{PGL}_3(\mathbb{C})$.
- Let $\kappa \in {}^L G$ be the element represented by

$$\kappa = \begin{pmatrix} 1 & & \\ & \zeta_3 & \\ & & \zeta_3^2 \end{pmatrix}, \quad \zeta_3^3 = 1.$$

- Let ${}^L H = \mathrm{Cent}_{{}^L G}(\kappa)$, generated by the diagonal torus and the permutation $\sigma_H = (123)$.

A Special Case: SL_3

- Let $G = \mathrm{SL}_3$ and ${}^L G = \mathrm{PGL}_3(\mathbb{C})$.
- Let $\kappa \in {}^L G$ be the element represented by

$$\kappa = \begin{pmatrix} 1 & & \\ & \zeta_3 & \\ & & \zeta_3^2 \end{pmatrix}, \quad \zeta_3^3 = 1.$$

- Let ${}^L H = \mathrm{Cent}_{{}^L G}(\kappa)$, generated by the diagonal torus and the permutation $\sigma_H = (123)$.
- Consider $V = \mathfrak{sl}_3(\mathbb{C})$ the adjoint representation of ${}^L G$, and $V[0]$ the 0-weight subspace.

A Special Case: SL_3

- Let $G = \mathrm{SL}_3$ and ${}^L G = \mathrm{PGL}_3(\mathbb{C})$.
- Let $\kappa \in {}^L G$ be the element represented by

$$\kappa = \begin{pmatrix} 1 & & \\ & \zeta_3 & \\ & & \zeta_3^2 \end{pmatrix}, \quad \zeta_3^3 = 1.$$

- Let ${}^L H = \mathrm{Cent}_{{}^L G}(\kappa)$, generated by the diagonal torus and the permutation $\sigma_H = (123)$.
- Consider $V = \mathfrak{sl}_3(\mathbb{C})$ the adjoint representation of ${}^L G$, and $V[0]$ the 0-weight subspace.
- Let $\check{\varpi}_1$, $\check{\varpi}_2$ be the fundamental coweights of G.

A Special Case: SL_3

- Let $G = \mathrm{SL}_3$ and ${}^L G = \mathrm{PGL}_3(\mathbb{C})$.
- Let $\kappa \in {}^L G$ be the element represented by

$$\kappa = \begin{pmatrix} 1 & & \\ & \zeta_3 & \\ & & \zeta_3^2 \end{pmatrix}, \quad \zeta_3^3 = 1.$$

- Let ${}^L H = \mathrm{Cent}_{{}^L G}(\kappa)$, generated by the diagonal torus and the permutation $\sigma_H = (123)$.
- Consider $V = \mathfrak{sl}_3(\mathbb{C})$ the adjoint representation of ${}^L G$, and $V[0]$ the 0-weight subspace.
- Let $\check{\varpi}_1, \check{\varpi}_2$ be the fundamental coweights of G.

Theorem (Example of AsFL)

The eigenvalues of σ_H on $V[0]$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad i = 1, 2.$$

Verification

- $\check{\varpi}_1$ may be represented by the character on the diagonal torus of $\mathrm{GL}_3(\mathbb{C})$:

$$\check{\varpi}_1 : \begin{pmatrix} t_1 & & \\ & t_2 & \\ & & t_3 \end{pmatrix} \longmapsto t_1,$$

and $\check{\varpi}_2$ is

$$\check{\varpi}_2 : \begin{pmatrix} t_1 & & \\ & t_2 & \\ & & t_3 \end{pmatrix} \longmapsto t_1 t_2.$$

Verification

- $\check{\varpi}_1$ may be represented by the character on the diagonal torus of $\mathrm{GL}_3(\mathbb{C})$:

$$\check{\varpi}_1 : \begin{pmatrix} t_1 & & \\ & t_2 & \\ & & t_3 \end{pmatrix} \longmapsto t_1,$$

  and $\check{\varpi}_2$ is

$$\check{\varpi}_2 : \begin{pmatrix} t_1 & & \\ & t_2 & \\ & & t_3 \end{pmatrix} \longmapsto t_1 t_2.$$

- Neither $\langle \kappa, \check{\varpi}_1 \rangle$ nor $\langle \kappa, \check{\varpi}_2 \rangle$ makes sense since $\kappa \in \mathrm{PGL}_3$, but $\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle$ makes sense for both i.

Verification

- $\check{\varpi}_1$ may be represented by the character on the diagonal torus of $\mathrm{GL}_3(\mathbb{C})$:

$$\check{\varpi}_1 : \begin{pmatrix} t_1 & & \\ & t_2 & \\ & & t_3 \end{pmatrix} \longmapsto t_1,$$

  and $\check{\varpi}_2$ is

$$\check{\varpi}_2 : \begin{pmatrix} t_1 & & \\ & t_2 & \\ & & t_3 \end{pmatrix} \longmapsto t_1 t_2.$$

- Neither $\langle \kappa, \check{\varpi}_1 \rangle$ nor $\langle \kappa, \check{\varpi}_2 \rangle$ makes sense since $\kappa \in \mathrm{PGL}_3$, but $\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle$ makes sense for both i.
- The two pairings are ζ_3 and ζ_3^2, which are precisely the eigenvalues of σ_H on the Cartan of $\mathfrak{sl}_3(\mathbb{C})$.

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \leq i \leq n)$, then σ_H is the permutation $(12 \cdots n)$.

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \le i \le n)$, then σ_H is the permutation $(12 \cdots n)$.
- Eigenvalues of σ_H on the Cartan $V[0]$ of $V = \mathfrak{sl}_n(\mathbb{C})$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad 1 \le i \le n-1.$$

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \le i \le n)$, then σ_H is the permutation $(12 \cdots n)$.
- Eigenvalues of σ_H on the Cartan $V[0]$ of $V = \mathfrak{sl}_n(\mathbb{C})$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad 1 \le i \le n-1.$$

- More generally, the above formula holds when:

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \le i \le n)$, then σ_H is the permutation $(12 \cdots n)$.
- Eigenvalues of σ_H on the Cartan $V[0]$ of $V = \mathfrak{sl}_n(\mathbb{C})$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad 1 \le i \le n-1.$$

- More generally, the above formula holds when:
  - G is split, semisimple and simply-connected;

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \le i \le n)$, then σ_H is the permutation $(12 \cdots n)$.
- Eigenvalues of σ_H on the Cartan $V[0]$ of $V = \mathfrak{sl}_n(\mathbb{C})$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad 1 \le i \le n-1.$$

- More generally, the above formula holds when:
  - G is split, semisimple and simply-connected;
  - κ is part of an unramified endoscopic datum (H, κ, ξ);

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \le i \le n)$, then σ_H is the permutation $(12 \cdots n)$.
- Eigenvalues of σ_H on the Cartan $V[0]$ of $V = \mathfrak{sl}_n(\mathbb{C})$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad 1 \le i \le n-1.$$

- More generally, the above formula holds when:
  - G is split, semisimple and simply-connected;
  - κ is part of an unramified endoscopic datum (H, κ, ξ);
  - $V = \check{\mathfrak{g}}(\mathbb{C})$ is the adjoint representation of ${}^L G$.

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \le i \le n)$, then σ_H is the permutation $(12 \cdots n)$.
- Eigenvalues of σ_H on the Cartan $V[0]$ of $V = \mathfrak{sl}_n(\mathbb{C})$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad 1 \le i \le n-1.$$

- More generally, the above formula holds when:
  - G is split, semisimple and simply-connected;
  - κ is part of an unramified endoscopic datum (H, κ, ξ);
  - $V = \check{\mathfrak{g}}(\mathbb{C})$ is the adjoint representation of ${}^L G$.
- It resembles some old computation by Hales (1993, *A Simple Definition of Transfer Factors for Unramified Groups*).

Generalizations

- When $G = \mathrm{SL}_n$, ${}^L G = \mathrm{PGL}_n(\mathbb{C})$, and κ is represented by a diagonal matrix with entries ζ_n^i $(1 \le i \le n)$, then σ_H is the permutation $(12 \cdots n)$.
- Eigenvalues of σ_H on the Cartan $V[0]$ of $V = \mathfrak{sl}_n(\mathbb{C})$ are precisely

$$\langle \kappa, \check{\varpi}_i - \sigma_H(\check{\varpi}_i) \rangle, \quad 1 \le i \le n-1.$$

- More generally, the above formula holds when:
  - G is split, semisimple and simply-connected;
  - κ is part of an unramified endoscopic datum (H, κ, ξ);
  - $V = \check{\mathfrak{g}}(\mathbb{C})$ is the adjoint representation of ${}^L G$.
- It resembles some old computation by Hales (1993, *A Simple Definition of Transfer Factors for Unramified Groups*).

Question

What about general (unramified) G, representation V, and weight space $V[\mu]$ where μ is not necessarily 0?

Asymptotic FL for 0-weights

- Let F be a local field, $\mathcal{O}$ its valuation ring. Let σ be a Frobenius. Let G be a reductive group over $\mathcal{O}$, (H, κ, ξ) an unramified endoscopic datum.

Asymptotic FL for 0-weights

- Let F be a local field, $\mathcal{O}$ its valuation ring. Let σ be a Frobenius. Let G be a reductive group over $\mathcal{O}$, (H, κ, ξ) an unramified endoscopic datum.
- Let V be an algebraic representation of ${}^L G = \check{\mathbf{G}} \rtimes \langle \sigma \rangle$.

Asymptotic FL for 0-weights

- Let F be a local field, $\mathcal{O}$ its valuation ring. Let σ be a Frobenius. Let G be a reductive group over $\mathcal{O}$, (H, κ, ξ) an unramified endoscopic datum.
- Let V be an algebraic representation of ${}^L G = \check{\mathbf{G}} \rtimes \langle \sigma \rangle$.
- It induces a representation $V_{H,\xi}$ of ${}^L H = \check{\mathbf{H}} \rtimes \langle \sigma_H \rangle$ through $\xi \colon {}^L H \to {}^L G$.

Asymptotic FL for 0-weights

- Let F be a local field, $\mathcal{O}$ its valuation ring. Let σ be a Frobenius. Let G be a reductive group over $\mathcal{O}$, (H, κ, ξ) an unramified endoscopic datum.
- Let V be an algebraic representation of ${}^L G = \check{\mathbf{G}} \rtimes \langle \sigma \rangle$.
- It induces a representation $V_{H,\xi}$ of ${}^L H = \check{\mathbf{H}} \rtimes \langle \sigma_H \rangle$ through $\xi \colon {}^L H \to {}^L G$.
- The notations of the following theorem will be explained in the next few slides:

Asymptotic FL for 0-weights

- Let F be a local field, $\mathcal{O}$ its valuation ring. Let σ be a Frobenius. Let G be a reductive group over $\mathcal{O}$, (H, κ, ξ) an unramified endoscopic datum.
- Let V be an algebraic representation of ${}^L G = \check{\mathbf{G}} \rtimes \langle \sigma \rangle$.
- It induces a representation $V_{H,\xi}$ of ${}^L H = \check{\mathbf{H}} \rtimes \langle \sigma_H \rangle$ through $\xi \colon {}^L H \to {}^L G$.
- The notations of the following theorem will be explained in the next few slides:

Theorem (AsFL for 0-weight, W 2024)

We have an isomorphism of Frobenius modules

$$\mathfrak{v}[0]_\kappa \cong \mathfrak{v}_{H,\xi}[0]_{\mathrm{st}}.$$

Asymptotic FL for 0-weights

- Let F be a local field, $\mathcal{O}$ its valuation ring. Let σ be a Frobenius. Let G be a reductive group over $\mathcal{O}$, (H, κ, ξ) an unramified endoscopic datum.
- Let V be an algebraic representation of ${}^L G = \check{\mathbf{G}} \rtimes \langle \sigma \rangle$.
- It induces a representation $V_{H,\xi}$ of ${}^L H = \check{\mathbf{H}} \rtimes \langle \sigma_H \rangle$ through $\xi \colon {}^L H \to {}^L G$.
- The notations of the following theorem will be explained in the next few slides:

Theorem (AsFL for 0-weight, W 2024)

We have an isomorphism of Frobenius modules

$$\mathfrak{v}[0]_\kappa \cong \mathfrak{v}_{H,\xi}[0]_{\mathrm{st}}.$$

- As vector spaces, $\mathfrak{v}[0]_\kappa$ is isomorphic to the 0-weight subspace $V[0]$. Similarly for $\mathfrak{v}_{H,\xi}[0]_{\mathrm{st}}$.

Asymptotic FL for 0-weights

- We have the canonical decomposition

$$V = \bigoplus_{\lambda} V_\lambda \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V),$$

where V_λ are highest-weight representations of $\check{\mathbf{G}}$, with normalized σ-actions.

Asymptotic FL for 0-weights

- We have the canonical decomposition

$$V = \bigoplus_{\lambda} V_\lambda \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V),$$

  where V_λ are highest-weight representations of $\check{\mathbf{G}}$, with normalized σ-actions.
- Let $\mathscr{B}_\lambda$ be the canonical Kashiwara crystal (discussed later) associated with V_λ.

Asymptotic FL for 0-weights

- We have the canonical decomposition

$$V = \bigoplus_{\lambda} V_\lambda \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V),$$

  where V_λ are highest-weight representations of $\check{\mathbf{G}}$, with normalized σ-actions.
- Let $\mathscr{B}_\lambda$ be the canonical Kashiwara crystal (discussed later) associated with V_λ.
- May view $\mathscr{B}_\lambda$ as a particularly well-chosen basis of V_λ consisting of weight vectors.

Asymptotic FL for 0-weights

- We have the canonical decomposition

$$V = \bigoplus_{\lambda} V_\lambda \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V),$$

  where V_λ are highest-weight representations of $\check{\mathbf{G}}$, with normalized σ-actions.
- Let $\mathscr{B}_\lambda$ be the canonical Kashiwara crystal (discussed later) associated with V_λ.
- May view $\mathscr{B}_\lambda$ as a particularly well-chosen basis of V_λ consisting of weight vectors.
- To each element $x \in \mathscr{B}_\lambda$ and any simple coroot $\check{\alpha}_i$, there are two canonically associated numbers $\epsilon_i(x)$ and $\phi_i(x)$:

$$\epsilon_i(x) = \max\{n \geq 0 \mid e_i^n(x) \neq 0\}, \quad \phi_i(x) = \max\{n \geq 0 \mid f_i^n(x) \neq 0\}.$$

  e_i and f_i are nilpotent operators in the $\mathfrak{sl}_2$-triple of $\check{\alpha}_i$.

Asymptotic FL for 0-weights

- We have the canonical decomposition

$$V = \bigoplus_{\lambda} V_\lambda \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V),$$

  where V_λ are highest-weight representations of $\check{\mathbf{G}}$, with normalized σ-actions.
- Let $\mathscr{B}_\lambda$ be the canonical Kashiwara crystal (discussed later) associated with V_λ.
- May view $\mathscr{B}_\lambda$ as a particularly well-chosen basis of V_λ consisting of weight vectors.
- To each element $x \in \mathscr{B}_\lambda$ and any simple coroot $\check{\alpha}_i$, there are two canonically associated numbers $\epsilon_i(x)$ and $\phi_i(x)$:

$$\epsilon_i(x) = \max\{n \geq 0 \mid e_i^n(x) \neq 0\}, \quad \phi_i(x) = \max\{n \geq 0 \mid f_i^n(x) \neq 0\}.$$

  e_i and f_i are nilpotent operators in the $\mathfrak{sl}_2$-triple of $\check{\alpha}_i$.
- If $x \in \mathscr{B}_\lambda[0]$ has weight 0, define

$$\rho_x = \sum_i \epsilon_i(x)\check{\varpi}_i.$$

Asymptotic FL for 0-weights

- We have the canonical decomposition

$$V = \bigoplus_{\lambda} V_\lambda \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V),$$

  where V_λ are highest-weight representations of $\check{\mathbf{G}}$, with normalized σ-actions.
- Let $\mathscr{B}_\lambda$ be the canonical Kashiwara crystal (discussed later) associated with V_λ.
- May view $\mathscr{B}_\lambda$ as a particularly well-chosen basis of V_λ consisting of weight vectors.
- To each element $x \in \mathscr{B}_\lambda$ and any simple coroot $\check{\alpha}_i$, there are two canonically associated numbers $\epsilon_i(x)$ and $\phi_i(x)$:

$$\epsilon_i(x) = \max\{n \geq 0 \mid e_i^n(x) \neq 0\}, \quad \phi_i(x) = \max\{n \geq 0 \mid f_i^n(x) \neq 0\}.$$

  e_i and f_i are nilpotent operators in the $\mathfrak{sl}_2$-triple of $\check{\alpha}_i$.
- If $x \in \mathscr{B}_\lambda[0]$ has weight 0, define

$$\rho_x = \sum_i \epsilon_i(x)\check{\varpi}_i.$$

- We use $\mathfrak{V}_\lambda[\mu]$ to denote the $\mathbb{C}$-vector space spanned by elements in $\mathscr{B}_\lambda[\mu]$.

Asymptotic FL for 0-weights

- The vector space

$$\mathfrak{v}[0] := \bigoplus_{\lambda} \mathfrak{v}_\lambda[0] \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V)$$

carries a natural σ-action. Denote this σ-module by $\mathfrak{v}[0]_{\mathrm{st}}$.

Asymptotic FL for 0-weights

- The vector space

$$\mathfrak{V}[0] := \bigoplus_{\lambda} \mathfrak{V}_\lambda[0] \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V)$$

  carries a natural σ-action. Denote this σ-module by $\mathfrak{V}[0]_{\mathrm{st}}$.
- We define another κ-twisted action of σ on $\mathfrak{V}[0]$ by

$$x \longmapsto \langle \kappa, \sigma(\rho_x) - \sigma_H(\rho_x) \rangle \sigma(x),$$

  and we denote the σ-module structure by $\mathfrak{V}[0]_\kappa$.

Asymptotic FL for 0-weights

- The vector space

$$\mathfrak{V}[0] := \bigoplus_{\lambda} \mathfrak{V}_\lambda[0] \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V)$$

carries a natural σ-action. Denote this σ-module by $\mathfrak{V}[0]_{\mathrm{st}}$.

- We define another κ-twisted action of σ on $\mathfrak{V}[0]$ by

$$x \longmapsto \langle \kappa, \sigma(\rho_x) - \sigma_H(\rho_x) \rangle \sigma(x),$$

and we denote the σ-module structure by $\mathfrak{V}[0]_\kappa$.

- Similarly, using Kashiwara crystals of $\check{\mathbf{H}}$, we have a vector space

$$\mathfrak{V}_{H,\xi}[0] = \bigoplus_{\lambda_H} \mathfrak{V}_{H,\lambda_H}[0] \otimes \mathrm{Hom}_{\check{\mathbf{H}}}(V_{H,\lambda_H}, V_{H,\xi}),$$

as well as the induced σ_H-module $\mathfrak{V}_{H,\xi}[0]_{\mathrm{st}}$.

Asymptotic FL for 0-weights

- The vector space

$$\mathfrak{V}[0] := \bigoplus_{\lambda} \mathfrak{V}_\lambda[0] \otimes \mathrm{Hom}_{\check{\mathbf{G}}}(V_\lambda, V)$$

carries a natural σ-action. Denote this σ-module by $\mathfrak{V}[0]_{\mathrm{st}}$.

- We define another κ-twisted action of σ on $\mathfrak{V}[0]$ by

$$x \longmapsto \langle \kappa, \sigma(\rho_x) - \sigma_H(\rho_x) \rangle \sigma(x),$$

and we denote the σ-module structure by $\mathfrak{V}[0]_\kappa$.

- Similarly, using Kashiwara crystals of $\check{\mathbf{H}}$, we have a vector space

$$\mathfrak{V}_{H,\xi}[0] = \bigoplus_{\lambda_H} \mathfrak{V}_{H,\lambda_H}[0] \otimes \mathrm{Hom}_{\check{\mathbf{H}}}(V_{H,\lambda_H}, V_{H,\xi}),$$

as well as the induced σ_H-module $\mathfrak{V}_{H,\xi}[0]_{\mathrm{st}}$.

- AsFL claims that $\mathfrak{V}[0]_\kappa \cong \mathfrak{V}_{H,\xi}[0]_{\mathrm{st}}$, as Frobenius modules.

Remarks on Asymptotic FL

Remarks on Asymptotic FL

General weights

- The statement for general weights μ is a lot more complicated.

Remarks on Asymptotic FL

General weights

- The statement for general weights μ is a lot more complicated.
- It uses the canonical Weyl group action on crystals, and explicit pieces of the transfer factor also show up.

Remarks on Asymptotic FL

General weights

- The statement for general weights μ is a lot more complicated.
- It uses the canonical Weyl group action on crystals, and explicit pieces of the transfer factor also show up.

FL vs Asymptotic FL

Remarks on Asymptotic FL

General weights

- The statement for general weights μ is a lot more complicated.
- It uses the canonical Weyl group action on crystals, and explicit pieces of the transfer factor also show up.

FL vs Asymptotic FL

- Using multiplicative Hitchin fibrations, one can reduce FL to AsFL.

Remarks on Asymptotic FL

General weights

- The statement for general weights μ is a lot more complicated.
- It uses the canonical Weyl group action on crystals, and explicit pieces of the transfer factor also show up.

FL vs Asymptotic FL

- Using multiplicative Hitchin fibrations, one can reduce FL to AsFL.
- The Lie-AsFL (implicit in Ngô's work), on the other hand, is trivial:

$$\mathbb{C} \cong \mathbb{C},$$

both being trivial Frobenius modules, thanks to Kottwitz (1999, *Transfer Factors for Lie Algebras*).

The Transfer Factor

1 Motivating example: SL_2
2 From FL to AsFL
3 On Top of Mindolluin
4 Asymptotic Fundamental Lemma
5 The Transfer Factor
6 Kashiwara Crystals

Components of the Transfer Factor

- The original Langlands-Shelstad transfer factor is a product of 5 components:

$$\Delta = \Delta_{\mathrm{I}}\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_1}\Delta_{\mathrm{III}_2}\Delta_{\mathrm{IV}}.$$

Components of the Transfer Factor

- The original Langlands-Shelstad transfer factor is a product of 5 components:

$$\Delta = \Delta_{\mathrm{I}}\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_1}\Delta_{\mathrm{III}_2}\Delta_{\mathrm{IV}}.$$

- Some components depend on auxiliary choices, including those called a-data and χ-data, but the product Δ is independent of such choices.

Components of the Transfer Factor

- The original Langlands-Shelstad transfer factor is a product of 5 components:

$$\Delta = \Delta_{\mathrm{I}}\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_1}\Delta_{\mathrm{III}_2}\Delta_{\mathrm{IV}}.$$

- Some components depend on auxiliary choices, including those called a-data and χ-data, but the product Δ is independent of such choices.
- The term Δ_{IV} is just Tate twists, so we ignore it.

Components of the Transfer Factor

- The original Langlands-Shelstad transfer factor is a product of 5 components:

$$\Delta = \Delta_{\mathrm{I}}\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_1}\Delta_{\mathrm{III}_2}\Delta_{\mathrm{IV}}.$$

- Some components depend on auxiliary choices, including those called a-data and χ-data, but the product Δ is independent of such choices.
- The term Δ_{IV} is just Tate twists, so we ignore it.
- The term Δ_{III_1} is used to define $\mathbf{O}^{\kappa}_{\gamma}$, which does not depend on a-data nor χ-data, hence relatively easy to handle.

Components of the Transfer Factor

- The original Langlands-Shelstad transfer factor is a product of 5 components:

$$\Delta = \Delta_{\mathrm{I}}\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_1}\Delta_{\mathrm{III}_2}\Delta_{\mathrm{IV}}.$$

- Some components depend on auxiliary choices, including those called a-data and χ-data, but the product Δ is independent of such choices.
- The term Δ_{IV} is just Tate twists, so we ignore it.
- The term Δ_{III_1} is used to define $\mathbf{O}^{\kappa}_{\gamma}$, which does not depend on a-data nor χ-data, hence relatively easy to handle.
- We focus on the remaining terms: Δ_{I}, Δ_{II} and Δ_{III_2}.

The Term Δ_{I}

- Recall $F = k((\pi))$ and $\mathcal{O} = k[[\pi]]$. Let $T \subset G$ be the maximal torus in an F-pinning of G.

The Term Δ_{I}

- Recall $F = k((\pi))$ and $\mathcal{O} = k[[\pi]]$. Let $T \subset G$ be the maximal torus in an F-pinning of G.
- Let $\gamma \in G(F)$ be strongly regular semisimple and unramified, matching $\gamma_H \in H(F)$.

The Term Δ_{I}

- Recall $F = k((\pi))$ and $\mathcal{O} = k[[\pi]]$. Let $T \subset G$ be the maximal torus in an F-pinning of G.
- Let $\gamma \in G(F)$ be strongly regular semisimple and unramified, matching $\gamma_H \in H(F)$.
- Let $\breve{F} = \bar{k}((\pi))$ be the unramified closure of F. Then we can find some element

$$t \in T(\breve{F}),$$
$$\tilde{\sigma}' \coloneqq \tilde{w} \rtimes \sigma \in \mathrm{N}_G(T)(\breve{F}) \rtimes \sigma, \quad \text{(twisted Frobenius)}$$

such that t is conjugate to γ and $\tilde{\sigma}'(t) = t$.

The Term Δ_{I}

- Recall $F = k((\pi))$ and $\mathcal{O} = k[[\pi]]$. Let $T \subset G$ be the maximal torus in an F-pinning of G.
- Let $\gamma \in G(F)$ be strongly regular semisimple and unramified, matching $\gamma_H \in H(F)$.
- Let $\breve{F} = \bar{k}((\pi))$ be the unramified closure of F. Then we can find some element

$$t \in T(\breve{F}),$$
$$\tilde{\sigma}' \coloneqq \tilde{w} \rtimes \sigma \in \mathrm{N}_G(T)(\breve{F}) \rtimes \sigma, \quad \text{(twisted Frobenius)}$$

  such that t is conjugate to γ and $\tilde{\sigma}'(t) = t$.
- Write $\tilde{w}$ as $\mathbf{cn}(w)$ where $\mathbf{n}(w)$ is given by the Tits section $W \to \mathrm{N}_G(T)$.

The Term Δ_{I}

- Recall $F = k((\pi))$ and $\mathcal{O} = k[[\pi]]$. Let $T \subset G$ be the maximal torus in an F-pinning of G.
- Let $\gamma \in G(F)$ be strongly regular semisimple and unramified, matching $\gamma_H \in H(F)$.
- Let $\breve{F} = \bar{k}((\pi))$ be the unramified closure of F. Then we can find some element

$$t \in T(\breve{F}),$$
$$\tilde{\sigma}' \coloneqq \tilde{w} \rtimes \sigma \in \mathrm{N}_G(T)(\breve{F}) \rtimes \sigma, \quad \text{(twisted Frobenius)}$$

  such that t is conjugate to γ and $\tilde{\sigma}'(t) = t$.
- Write $\tilde{w}$ as $\mathbf{c}\mathbf{n}(w)$ where $\mathbf{n}(w)$ is given by the Tits section $W \to \mathrm{N}_G(T)$.
- The Galois cocycle $\sigma \mapsto \mathbf{c}$ can be paired with κ, denoted by $\kappa(\mathbf{c})$.

The Term Δ_{I}

- Recall $F = k((\pi))$ and $\mathcal{O} = k[[\pi]]$. Let $T \subset G$ be the maximal torus in an F-pinning of G.
- Let $\gamma \in G(F)$ be strongly regular semisimple and unramified, matching $\gamma_H \in H(F)$.
- Let $\breve{F} = \bar{k}((\pi))$ be the unramified closure of F. Then we can find some element

$$t \in T(\breve{F}),$$
$$\tilde{\sigma}' \coloneqq \tilde{w} \rtimes \sigma \in \mathrm{N}_G(T)(\breve{F}) \rtimes \sigma, \quad \text{(twisted Frobenius)}$$

  such that t is conjugate to γ and $\tilde{\sigma}'(t) = t$.
- Write $\tilde{w}$ as $\mathbf{c}\mathbf{n}(w)$ where $\mathbf{n}(w)$ is given by the Tits section $W \to \mathrm{N}_G(T)$.
- The Galois cocycle $\sigma \mapsto \mathbf{c}$ can be paired with κ, denoted by $\kappa(\mathbf{c})$.
- Δ_{I} depends on the choice of a-data, and when γ is unramified, there is an "obvious" choice, with which we have

$$\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}.$$

Components of MASF

- Orbital integrals of y are geometrized by multiplicative affine Springer fibers (MASF).

Components of MASF

- Orbital integrals of γ are geometrized by multiplicative affine Springer fibers (MASF).
- J. Chi (2019) showed that the geometric irreducible components of the MASF:

$$\mathcal{M}_\gamma^\lambda := \left\{ g \in G(\breve{F})/G(\breve{\mathcal{O}}) \;\middle|\; g^{-1}\gamma g \in G(\breve{\mathcal{O}})\pi^\lambda G(\breve{\mathcal{O}}) \right\}$$

are in natural bijection with

$$\mathrm{Irr}(S_\gamma^\lambda) \times \breve{\mathbb{X}},$$

where

Components of MASF

- Orbital integrals of γ are geometrized by multiplicative affine Springer fibers (MASF).
- J. Chi (2019) showed that the geometric irreducible components of the MASF:

$$\mathcal{M}_\gamma^\lambda := \left\{ g \in G(\breve{F})/G(\breve{\mathcal{O}}) \;\middle|\; g^{-1}\gamma g \in G(\breve{\mathcal{O}})\pi^\lambda G(\breve{\mathcal{O}}) \right\}$$

are in natural bijection with

$$\mathrm{Irr}(S_\gamma^\lambda) \times \check{\mathbb{X}},$$

where

  - $\check{\mathbb{X}} \simeq T(\breve{F})/T(\breve{\mathcal{O}})$ is the cocharacter lattice;

Components of MASF

- Orbital integrals of γ are geometrized by multiplicative affine Springer fibers (MASF).
- J. Chi (2019) showed that the geometric irreducible components of the MASF:

$$\mathcal{M}_\gamma^\lambda := \left\{ g \in G(\breve{F})/G(\breve{\mathcal{O}}) \;\middle|\; g^{-1}\gamma g \in G(\breve{\mathcal{O}})\pi^\lambda G(\breve{\mathcal{O}}) \right\}$$

are in natural bijection with

$$\mathrm{Irr}(S_\nu^\lambda) \times \breve{\mathbb{X}},$$

where

  - $\breve{\mathbb{X}} \simeq T(\breve{F})/T(\breve{\mathcal{O}})$ is the cocharacter lattice;
  - $\nu \in \breve{\mathbb{X}}$ is such that $t \in \pi^\nu T(\breve{\mathcal{O}})$;

Components of MASF

- Orbital integrals of γ are geometrized by multiplicative affine Springer fibers (MASF).
- J. Chi (2019) showed that the geometric irreducible components of the MASF:

$$\mathcal{M}_\gamma^\lambda := \left\{ g \in G(\breve{F})/G(\breve{\mathcal{O}}) \;\middle|\; g^{-1}\gamma g \in G(\breve{\mathcal{O}})\pi^\lambda G(\breve{\mathcal{O}}) \right\}$$

are in natural bijection with

$$\mathrm{Irr}(S_\nu^\lambda) \times \breve{\mathbb{X}},$$

where

  - $\breve{\mathbb{X}} \simeq T(\breve{F})/T(\breve{\mathcal{O}})$ is the cocharacter lattice;
  - $\nu \in \breve{\mathbb{X}}$ is such that $t \in \pi^\nu T(\breve{\mathcal{O}})$;
  - S_ν^λ is the Mirković-Vilonen (MV) cycle of type λ and weight ν.

Components of MASF

- Orbital integrals of γ are geometrized by multiplicative affine Springer fibers (MASF).
- J. Chi (2019) showed that the geometric irreducible components of the MASF:

$$\mathcal{M}_\gamma^\lambda := \left\{ g \in G(\breve{F})/G(\breve{\mathcal{O}}) \;\middle|\; g^{-1}\gamma g \in G(\breve{\mathcal{O}})\pi^\lambda G(\breve{\mathcal{O}}) \right\}$$

  are in natural bijection with

$$\mathrm{Irr}(S_\nu^\lambda) \times \breve{\mathbb{X}},$$

  where
  - $\breve{\mathbb{X}} \simeq T(\breve{F})/T(\breve{\mathcal{O}})$ is the cocharacter lattice;
  - $\nu \in \breve{\mathbb{X}}$ is such that $t \in \pi^\nu T(\breve{\mathcal{O}})$;
  - S_ν^λ is the Mirković-Vilonen (MV) cycle of type λ and weight ν.
- The question is then about the action of the twisted Frobenius $\tilde{\sigma}'$.

Twisted Frobenius Action

- We have seen that $\tilde{\sigma}' = \mathbf{cn}(w) \rtimes \sigma$. The effect of $\mathbf{c}$ is explained by the term $\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}$ (with the "obvious" a-datum).

Twisted Frobenius Action

- We have seen that $\tilde{\sigma}' = \mathbf{cn}(w) \rtimes \sigma$. The effect of $\mathbf{c}$ is explained by the term $\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}$ (with the "obvious" a-datum).
- Let $\sigma' = \mathbf{n}(w) \rtimes \sigma$.

Twisted Frobenius Action

- We have seen that $\tilde{\sigma}' = \mathbf{cn}(w) \rtimes \sigma$. The effect of $\mathbf{c}$ is explained by the term $\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}$ (with the "obvious" a-datum).
- Let $\sigma' = \mathbf{n}(w) \rtimes \sigma$.
- Let $(x, \theta) \in \mathrm{Irr}(S_\gamma^\lambda) \times \check{\mathbb{X}}(T)$ be an irreducible component of $\mathcal{M}_\gamma^\lambda$ and suppose

$$\sigma' \cdot (x, \theta) = (x', \theta'),$$

then how do we describe x' and θ' in terms of x and θ?

Twisted Frobenius Action

- We have seen that $\tilde{\sigma}' = \mathbf{cn}(w) \rtimes \sigma$. The effect of $\mathbf{c}$ is explained by the term $\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}$ (with the "obvious" a-datum).
- Let $\sigma' = \mathbf{n}(w) \rtimes \sigma$.
- Let $(\mathbf{x}, \theta) \in \mathrm{Irr}(S_\nu^\lambda) \times \check{\mathbb{X}}(T)$ be an irreducible component of $\mathcal{M}_\gamma^\lambda$ and suppose

$$\sigma' \cdot (\mathbf{x}, \theta) = (\mathbf{x}', \theta'),$$

  then how do we describe $\mathbf{x}'$ and θ' in terms of $\mathbf{x}$ and θ?
- Naive guess: $\mathrm{Irr}(S_\nu^\lambda)$ gives a natural basis of the weight space $V_\lambda[\nu]$ by geometric Satake, so maybe $\mathbf{x}' = \sigma'(\mathbf{x})$ as vectors in the representation V_λ?

Twisted Frobenius Action

- We have seen that $\tilde{\sigma}' = \mathbf{c}\mathbf{n}(w) \rtimes \sigma$. The effect of $\mathbf{c}$ is explained by the term $\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}$ (with the "obvious" a-datum).
- Let $\sigma' = \mathbf{n}(w) \rtimes \sigma$.
- Let $(x, \theta) \in \mathrm{Irr}(S_\nu^\lambda) \times \check{\mathbb{X}}(T)$ be an irreducible component of $\mathcal{M}_\gamma^\lambda$ and suppose

$$\sigma' \cdot (x, \theta) = (x', \theta'),$$

  then how do we describe x' and θ' in terms of x and θ?
- Naive guess: $\mathrm{Irr}(S_\nu^\lambda)$ gives a natural basis of the weight space $V_\lambda[\nu]$ by geometric Satake, so maybe $x' = \sigma'(x)$ as vectors in the representation V_λ?
- Problem one: σ' does not make sense as an element of ${}^L G$.

Twisted Frobenius Action

- We have seen that $\tilde{\sigma}' = \mathbf{cn}(w) \rtimes \sigma$. The effect of $\mathbf{c}$ is explained by the term $\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}$ (with the "obvious" a-datum).
- Let $\sigma' = \mathbf{n}(w) \rtimes \sigma$.
- Let $(\mathbf{x}, \theta) \in \mathrm{Irr}(S_\nu^\lambda) \times \check{\mathbb{X}}(T)$ be an irreducible component of $\mathcal{M}_\gamma^\lambda$ and suppose

$$\sigma' \cdot (\mathbf{x}, \theta) = (\mathbf{x}', \theta'),$$

  then how do we describe $\mathbf{x}'$ and θ' in terms of $\mathbf{x}$ and θ?
- Naive guess: $\mathrm{Irr}(S_\nu^\lambda)$ gives a natural basis of the weight space $V_\lambda[\nu]$ by geometric Satake, so maybe $\mathbf{x}' = \sigma'(\mathbf{x})$ as vectors in the representation V_λ?
- Problem one: σ' does not make sense as an element of ${}^L G$.
- Problem two: hard to make sense of the θ-part, nor the κ-twisting, using representations of L-groups.

Twisted Frobenius Action

- We have seen that $\tilde{\sigma}' = \mathbf{c}\mathbf{n}(w) \rtimes \sigma$. The effect of $\mathbf{c}$ is explained by the term $\Delta_{\mathrm{I}}(\gamma_H, \gamma) = \kappa(\mathbf{c})^{-1}$ (with the "obvious" a-datum).
- Let $\sigma' = \mathbf{n}(w) \rtimes \sigma$.
- Let $(\mathbf{x}, \theta) \in \mathrm{Irr}(S_\nu^\lambda) \times \check{\mathbb{X}}(T)$ be an irreducible component of $\mathcal{M}_\gamma^\lambda$ and suppose

$$\sigma' \cdot (\mathbf{x}, \theta) = (\mathbf{x}', \theta'),$$

  then how do we describe $\mathbf{x}'$ and θ' in terms of $\mathbf{x}$ and θ?
- Naive guess: $\mathrm{Irr}(S_\nu^\lambda)$ gives a natural basis of the weight space $V_\lambda[\nu]$ by geometric Satake, so maybe $\mathbf{x}' = \sigma'(\mathbf{x})$ as vectors in the representation V_λ?
- Problem one: σ' does not make sense as an element of ${}^L G$.
- Problem two: hard to make sense of the θ-part, nor the κ-twisting, using representations of L-groups.
- Correct answer: replace representations by (Kashiwara) crystals.

The Remaining Terms in Δ

- Braverman-Gaitsgory (2000): MV-cycles induce a crystal basis as well.

The Remaining Terms in Δ

- Braverman-Gaitsgory (2000): MV-cycles induce a crystal basis as well.
- The effect of σ' on the θ-part and the κ-twisting can be naturally explained using crystals.

The Remaining Terms in Δ

- Braverman-Gaitsgory (2000): MV-cycles induce a crystal basis as well.
- The effect of σ' on the θ-part and the κ-twisting can be naturally explained using crystals.
- The remaining terms $\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_2}$ can be seen as the compensation term (which is trivial when $\nu = 0$) resulting from the missing functor (explained later) in the following diagram:

$$\begin{tikzcd}
{}^L G\text{-reps} \arrow[r, "\xi^*"] \arrow[d, squiggly] & {}^L H\text{-reps} \arrow[d, squiggly] \\
{}^L G\text{-crystals} \arrow[r, dashed, "?"] & {}^L H\text{-crystals}
\end{tikzcd}.$$

The Remaining Terms in Δ

- Braverman-Gaitsgory (2000): MV-cycles induce a crystal basis as well.
- The effect of σ' on the θ-part and the κ-twisting can be naturally explained using crystals.
- The remaining terms $\Delta_{\mathrm{II}}\Delta_{\mathrm{III}_2}$ can be seen as the compensation term (which is trivial when $\nu = 0$) resulting from the missing functor (explained later) in the following diagram:

$$\begin{tikzcd}
{}^L G\text{-reps} \arrow[r, "\xi^*"] \arrow[d, squiggly] & {}^L H\text{-reps} \arrow[d, squiggly] \\
{}^L G\text{-crystals} \arrow[r, dashed, "?"] & {}^L H\text{-crystals}
\end{tikzcd}.$$

Wish

Is it possible to renormalize the transfer factor (including for ramified elements) using crystals, so that we can hope to formulate it categorically?

Kashiwara Crystals

1 Motivating example: SL_2
2 From FL to AsFL
3 On Top of Mindolluin
4 Asymptotic Fundamental Lemma
5 The Transfer Factor
6 Kashiwara Crystals

Kashiwara Crystals

- Highest-weight theory: the common theory of finite dimensional representations of compact Lie groups, algebraic reductive groups, or semisimple Lie algebras.

Kashiwara Crystals

- Highest-weight theory: the common theory of finite dimensional representations of compact Lie groups, algebraic reductive groups, or semisimple Lie algebras.
- Essentially built upon the theory of $\mathfrak{sl}_2$-triples given by simple roots.

Kashiwara Crystals

- Highest-weight theory: the common theory of finite dimensional representations of compact Lie groups, algebraic reductive groups, or semisimple Lie algebras.
- Essentially built upon the theory of $\mathfrak{sl}_2$-triples given by simple roots.
- Finite dimensional irreducible representations of $\mathfrak{sl}_2$ are easy: they are the spaces of homogeneous polynomials of degrees n:

$$\bigoplus_{i=0}^{n} \mathbb{C}x^i y^{n-i}.$$

The e-operator is $x\partial_y$ and the f-operator is $y\partial_x$.

Kashiwara Crystals

- Highest-weight theory: the common theory of finite dimensional representations of compact Lie groups, algebraic reductive groups, or semisimple Lie algebras.
- Essentially built upon the theory of $\mathfrak{sl}_2$-triples given by simple roots.
- Finite dimensional irreducible representations of $\mathfrak{sl}_2$ are easy: they are the spaces of homogeneous polynomials of degrees n:

$$\bigoplus_{i=0}^{n} \mathbb{C} x^i y^{n-i}.$$

  The e-operator is $x\partial_y$ and the f-operator is $y\partial_x$.
- Using the basis $x^i y^{n-i}$, one may visualize the representation as a line

$$\bullet \text{ — } \bullet \text{ — } \cdots \text{ — } \cdots \text{ — } \bullet \text{ — } \bullet$$

  with each vertex representing a monomial $x^i y^{n-i}$, and e and f takes one vertex to another, up to scaling.

- Ideally, one hopes for a similar result for arbitrary G.

Kashiwara Crystals

- Ideally, one hopes for a similar result for arbitrary G.
- Namely, a "good" basis of a representation, so that e_i and f_i given by simple roots take one vector to another, up to scaling.

Kashiwara Crystals

- Ideally, one hopes for a similar result for arbitrary G.
- Namely, a "good" basis of a representation, so that e_i and f_i given by simple roots take one vector to another, up to scaling.
- Too good to be true.

Kashiwara Crystals

- Ideally, one hopes for a similar result for arbitrary G.
- Namely, a "good" basis of a representation, so that e_i and f_i given by simple roots take one vector to another, up to scaling.
- Too good to be true.
- Next best thing: make it true in some "associated graded" sense.

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i : \mathscr{B} \to \mathbb{N}$ for each i,

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i : \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i : \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.
- One then defines the "associated graded" action of e_i on $\mathscr{B}$ by sending x to:

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i \colon \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.
- One then defines the "associated graded" action of e_i on $\mathscr{B}$ by sending x to:
  - 0, if $\epsilon_i(y) \leq \epsilon_i(x) - 1$ is always strict for all y with non-zero coefficients in $e_i(x)$;

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i \colon \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.
- One then defines the "associated graded" action of e_i on $\mathscr{B}$ by sending x to:
  - 0, if $\epsilon_i(y) \leq \epsilon_i(x) - 1$ is always strict for all y with non-zero coefficients in $e_i(x)$;
  - otherwise, the unique y in $e_i(x)$ such that $\epsilon_i(y) = \epsilon_i(x) - 1$.

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i \colon \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.
- One then defines the "associated graded" action of e_i on $\mathscr{B}$ by sending x to:
  - 0, if $\epsilon_i(y) \leq \epsilon_i(x) - 1$ is always strict for all y with non-zero coefficients in $e_i(x)$;
  - otherwise, the unique y in $e_i(x)$ such that $\epsilon_i(y) = \epsilon_i(x) - 1$.
- The graded f_i-action is the inverse of that of e_i.

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i \colon \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.
- One then defines the "associated graded" action of e_i on $\mathscr{B}$ by sending x to:
  - 0, if $\epsilon_i(y) \leq \epsilon_i(x) - 1$ is always strict for all y with non-zero coefficients in $e_i(x)$;
  - otherwise, the unique y in $e_i(x)$ such that $\epsilon_i(y) = \epsilon_i(x) - 1$.
- The graded f_i-action is the inverse of that of e_i.

The Miracle

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i \colon \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.
- One then defines the "associated graded" action of e_i on $\mathscr{B}$ by sending x to:
  - 0, if $\epsilon_i(y) \leq \epsilon_i(x) - 1$ is always strict for all y with non-zero coefficients in $e_i(x)$;
  - otherwise, the unique y in $e_i(x)$ such that $\epsilon_i(y) = \epsilon_i(x) - 1$.
- The graded f_i-action is the inverse of that of e_i.

The Miracle

- Not only does such $\mathscr{B}$ exist, they can be chosen for all representations compatible with tensor products.

Kashiwara Crystals

- More precisely, one can find a basis $\mathscr{B} \subset V$ consisting of weight vectors together with:
  - a partial order $\leq$ on $\mathscr{B}$, and
  - a grading function $\epsilon_i \colon \mathscr{B} \to \mathbb{N}$ for each i,

  such that for any $x \in \mathscr{B}$, $e_i(x) \in V$ is a linear combination of vectors $y \leq x$ with $\epsilon_i(y) \leq \epsilon_i(x) - 1$, and the equality is achieved for at most one y.
- One then defines the "associated graded" action of e_i on $\mathscr{B}$ by sending x to:
  - 0, if $\epsilon_i(y) \leq \epsilon_i(x) - 1$ is always strict for all y with non-zero coefficients in $e_i(x)$;
  - otherwise, the unique y in $e_i(x)$ such that $\epsilon_i(y) = \epsilon_i(x) - 1$.
- The graded f_i-action is the inverse of that of e_i.

The Miracle

- Not only does such $\mathscr{B}$ exist, they can be chosen for all representations compatible with tensor products.
- Any compatible system of such choices $\{\mathscr{B}\}$ are isomorphic (as collections of sets with e_i-operators and tensor products).

Crystal Graph

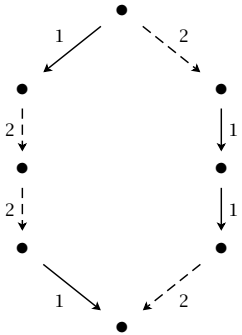
- Kashiwara crystals allow us to visualize the combinatorial essence of representations, in the form of crystal graphs.

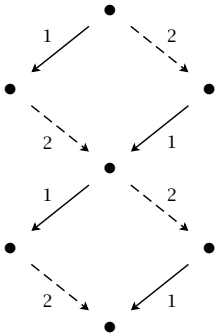
Crystal Graph

- Kashiwara crystals allow us to visualize the combinatorial essence of representations, in the form of crystal graphs.
- In fact, the crystals can be defined in an axiomatic way, and those coming from highest-weight representations correspond to so-called *normal crystals*.

Crystal Graph

- Kashiwara crystals allow us to visualize the combinatorial essence of representations, in the form of crystal graphs.
- In fact, the crystals can be defined in an axiomatic way, and those coming from highest-weight representations correspond to so-called *normal crystals*.
- Below are two crystals of SL_3. One is normal (adjoint representation), while the other is not:





Crystals vs Representations

- Crystals has many advantages over representations:

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.
  - Restrictions to standard Levi subgroups are extremely simple: simply delete the operators corresponding to the simple roots not in the Levi.

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.
  - Restrictions to standard Levi subgroups are extremely simple: simply delete the operators corresponding to the simple roots not in the Levi.
  - etc.

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.
  - Restrictions to standard Levi subgroups are extremely simple: simply delete the operators corresponding to the simple roots not in the Levi.
  - etc.
- Many disadvantages are also obvious:

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.
  - Restrictions to standard Levi subgroups are extremely simple: simply delete the operators corresponding to the simple roots not in the Levi.
  - etc.
- Many disadvantages are also obvious:
  - Tensor product is simple to describe on the level of basis elements x, but very complicated on the level of e_i and f_i operators.

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.
  - Restrictions to standard Levi subgroups are extremely simple: simply delete the operators corresponding to the simple roots not in the Levi.
  - etc.
- Many disadvantages are also obvious:
  - Tensor product is simple to describe on the level of basis elements x, but very complicated on the level of e_i and f_i operators.
  - Non-Levi branching rules become extremely complicated (near non-existence so far).

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.
  - Restrictions to standard Levi subgroups are extremely simple: simply delete the operators corresponding to the simple roots not in the Levi.
  - etc.
- Many disadvantages are also obvious:
  - Tensor product is simple to describe on the level of basis elements x, but very complicated on the level of e_i and f_i operators.
  - Non-Levi branching rules become extremely complicated (near non-existence so far).
  - In particular, the AsFL can be viewed as the branching rule for subgroups "of endoscopic type".

Crystals vs Representations

- Crystals has many advantages over representations:
  - Purely combinatorial.
  - Has a Weyl group action.
  - Restrictions to standard Levi subgroups are extremely simple: simply delete the operators corresponding to the simple roots not in the Levi.
  - etc.
- Many disadvantages are also obvious:
  - Tensor product is simple to describe on the level of basis elements x, but very complicated on the level of e_i and f_i operators.
  - Non-Levi branching rules become extremely complicated (near non-existence so far).
  - In particular, the AsFL can be viewed as the branching rule for subgroups "of endoscopic type".
  - Branching rules in general: functoriality?

Some Philosophical Questions

One can dream

- Is it possible to formulate functoriality (and co.) in terms of crystals (perhaps with a more generalized definition) instead of representations?

Some Philosophical Questions

One can dream

- Is it possible to formulate functoriality (and co.) in terms of crystals (perhaps with a more generalized definition) instead of representations?
- Failing that, is it always possible to reduce functoriality questions to a combinatorial core expressed by some sort of crystals?

Some Philosophical Questions

One can dream

- Is it possible to formulate functoriality (and co.) in terms of crystals (perhaps with a more generalized definition) instead of representations?
- Failing that, is it always possible to reduce functoriality questions to a combinatorial core expressed by some sort of crystals?
- The proof of endoscopic FL shows that the second dream is at least reasonable.

Thank you!